Multi-Agent System: Theories and Applications

Lecture 6: Brief Descriptions of Main Problems

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Outline

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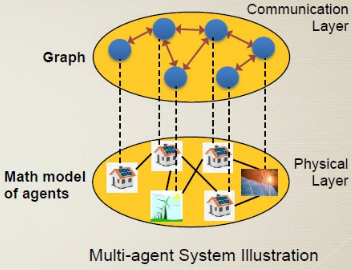
Formulation for Linear MASs

Assumption: Connectedness of $\mathcal{G}$

$$\dot{x}(t) = (I_N \otimes A)x(t) + (I_N \otimes B)u(t)$$
$$y(t) = (I_N \otimes C)x(t)$$
$$A \in \mathbb{R}^{n\times n}, B \in \mathbb{R}^{n\times m}, C \in \mathbb{R}^{p\times n}$$



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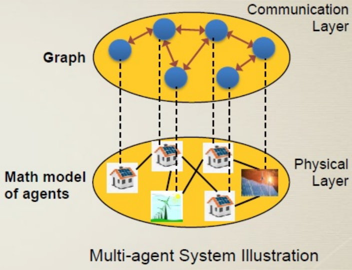

Agents are said to reach consensus if...

$\exists\, x^* \in \mathbb{R}^n$ s.t. the following is achieved ($t_f > 0$ is finite or $+\infty$),

$$\lim_{t\to t_f} x_i(t) = x^*,\ \forall\, i = 1, \ldots, N \Leftrightarrow \lim_{t\to t_f} x(t) = \mathbf{1}_N \otimes x^*$$

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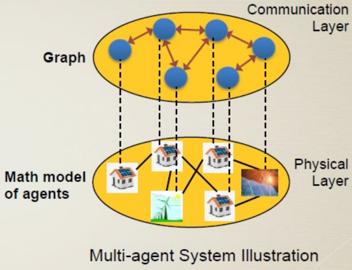

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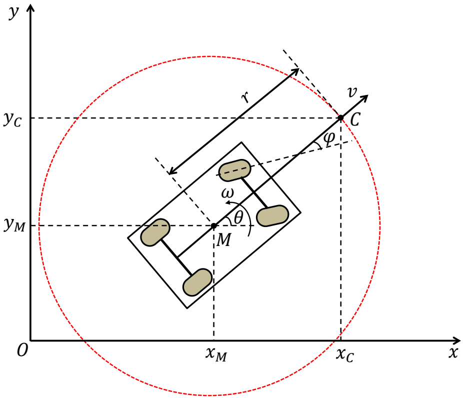
Question: What should be $u(t)$ to make $x(t)$ consensus?

Example: Consensus of A Vehicle Group

Mathematical model of vehicles:

$$\begin{aligned}
\dot{\theta}_i &= \omega_i = v_i \tan\varphi_i, \\
\dot{x}_{Mi} &= v_i \cos\theta_i, \\
\dot{y}_{Mi} &= v_i \sin\theta_i.
\end{aligned}$$



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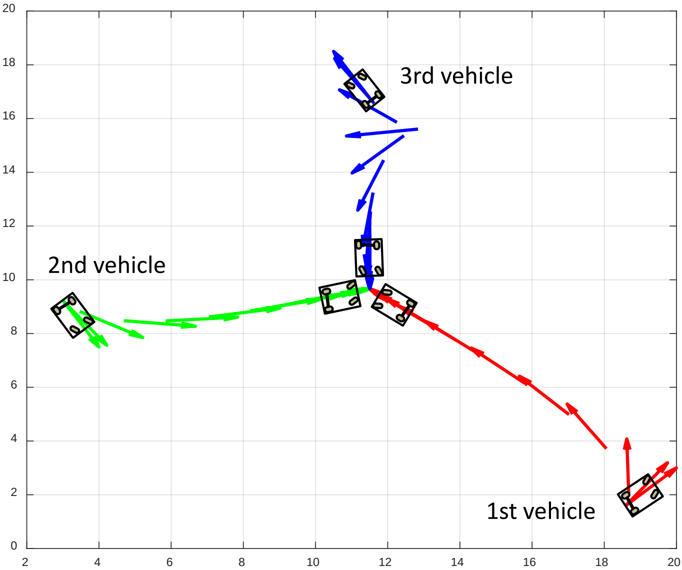
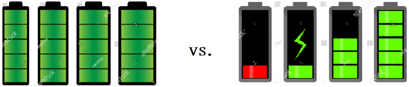


Other Applications

Consensus of batteries' levels



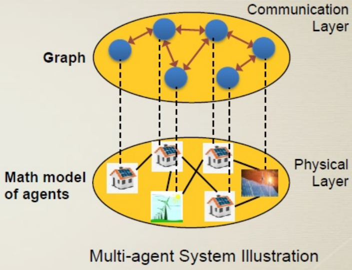
Social networks

Formulation

Assumption: Connectedness of $\mathcal{G}$

Each agent: an oscillator with

- natural frequency $\omega_i \in \mathbb{R}$
- phase $\theta_i(t) \in \mathbb{R}$
- frequency $\dot{\theta}_i(t) \in \mathbb{R}$



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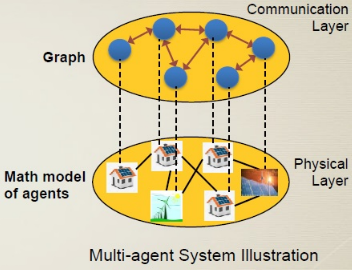

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$\exists\, \omega^* \in \mathbb{R}$ s.t. the following is achieved ($t_f > 0$ is finite or $+\infty$),

$$\lim_{t \to t_f} \dot{\theta}_i(t) = \omega^*,\ \forall\, i = 1, \ldots, N \Leftrightarrow \lim_{t \to t_f} \dot{\theta}(t) = \mathbf{1}_N \omega^*$$

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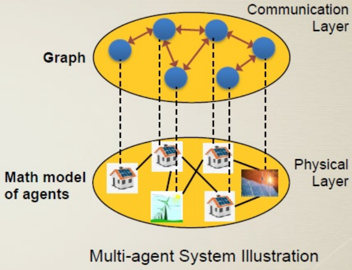

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Question: How oscillators are coupled to make their frequencies synchronized?

Applications in Electric Power Grids

- Kuramoto model:

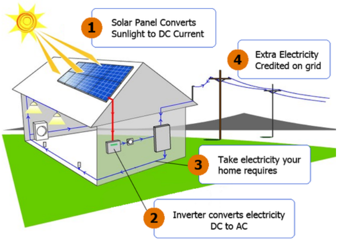
$$\dot{\theta}_i = \omega_i + \frac{K}{n}\sum_{j=1}^{n} \sin(\theta_j - \theta_i), \; i = 1, \ldots, n.$$

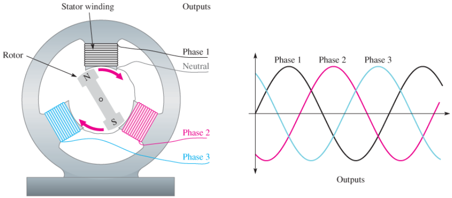
- Structure-preserving model for power network with synchronous generators:

$$M_i\ddot{\theta}_i + D_i\dot{\theta}_i = P_{m,i} - \sum_{j=1}^{n} a_{ij}\sin(\theta_i - \theta_j)$$

- Droop inverter-based microgrid with renewable generation:

$$D_i\dot{\delta}_i + \sum_{j=1}^{n} a_{ij}\sin(\delta_i - \delta_j) = P_i$$





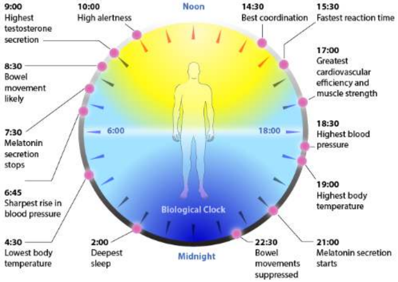
(a) Three-phase rotating-field generator (b) Three-phase sine wave

Applications in Biology

Goodwin-type model of oscillators of qth-order ($q \geq 3$),

$$\begin{cases} \dfrac{dx_1}{dt} &= f(x_q) - b_1 x_1, \quad f(x_q) = \dfrac{1}{1+x_q^p}, p > 0 : \text{Hill function}, \\ \dfrac{dx_2}{dt} &= x_1 - b_2 x_2, \\ & \vdots \\ \dfrac{dx_q}{dt} &= x_{q-1} - b_q x_q. \end{cases}$$

Circadian rhythm (c.f. Nobel Prize in Physiology or Medicine 2017)



Description for Linear Models of Vehicles

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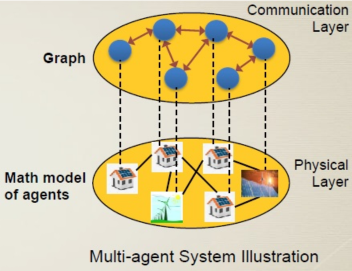

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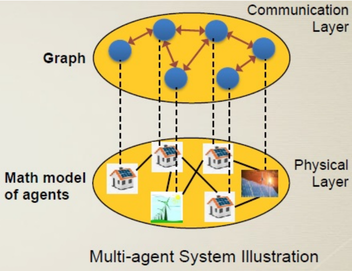

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(Aerial, ground, underwater, or water-surface) Vehicles are said to achieve a formation if...

$\exists\ x_{ij}^* \in \mathbb{R}^n, x_{ij}^* = -x_{ji}^*$ s.t. the following is held ($t_f > 0$ is finite or $+\infty$),

$$\lim_{t\to t_f} [x_i(t) - x_j(t)] = x_{ij}^*, i, j = 1, \ldots, N$$

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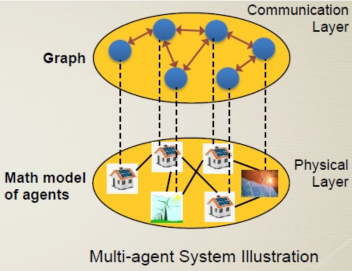

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Question: What should be $u(t)$ to make $x(t)$ achieve a desired formation?

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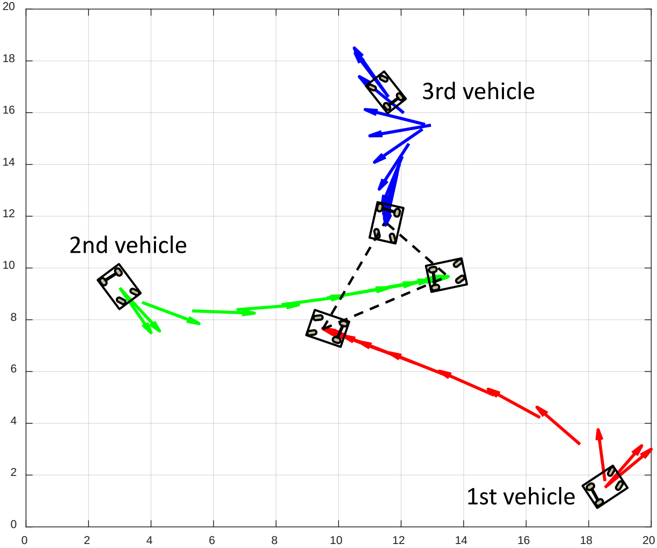
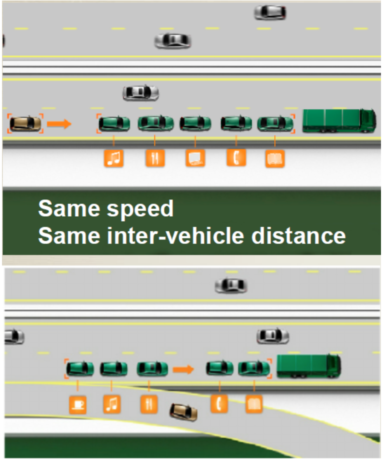
Example: Formation of An Unmanned Vehicle Group



Applications in Autonomous Vehicle Platoons



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Definition

Rendezvous is...

a consensus problem with **position-induced** interactions among agents, **with inter-agent collision avoidance.**

Important Points

- Agents have limited sensing and actuating abilities
- Each agent adjusts its move by observing its neighboring agents
- Agents avoid collision with other agents

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Definition

Flocking is...

similar to rendezvous but with not only inter-agent collision avoidance but also **obstacles avoidance.**

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Flocking is...

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Basic Principles

- Agents stay close to the others
- Agents tend to move in the same direction
- Agents avoid collision with other agents and with obstacles

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Applications in Vehicle Groups